Faculty of Liberal Arts and Professional Studies
YORK UNIVERSITY
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AP/ADMS 4504

Fixed Income Securities and Risk Management

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Group #	12

Assignment #2

Due Date: December 3, 2025

Group Work Statement

We, the undersigned:

- warrant that the work submitted herein is our group's work and not the work of others
- acknowledge that we have read and understood the Senate Policy on Academic Honesty
- acknowledge that it is a breach of the University Regulations to give and receive unauthorized assistance on a graded piece of work

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Question 1- Putable Bond and OAS

Question 1a.)- Was achieved by first constructing a binomial table. Once the binomial table was constructed, we had to calibrate the forward rates. This was done by following a consistent pattern. We first started by placing the par YTM of year zero in the first node of the binomial table as it is the current spot rate. We then used the forward rate of year 1 as a coupon for the payments of year 1 and the year ahead of year 1, in this case being year 2. We then set the value for year 2 to be par, 100 as to assist in calculating the year 1 price. Once we set the price for year 2, we multiplied both par values with the coupons added by 0.5 to give us the average price. We then divided the average price by the discount rate in year 1 to gain the price of year 1. We repeated this process with year zero. It is important to note that we only had the calibrated rate for year zero at this point, so we were effectively dividing the average price of the nodes in year 2 by 1+0. To solve the rate in year 1, we used the solver function to set the price in node zero to 100 to make the price have zero arbitrage. We then set forward rate of year 1 as the variable to change for node zero to equal 100, giving us a calibrated forward rate for year 1. Lastly, for every node in year 1, we altered the rate. For every vertical movement year 1, we multiplied the base forward rate by e^(n*2*volatility) with n equaling the number of vertical step-ups the node in question is from the base. We then repeated this process every following year to get the calibrated values.

Question 1b.)-To calculate the value of the put option, we insert the given face value and coupon values into their respective nodes. Once we do that, we use the calibrated forward rates from the prior question to derive the bond to its "z" price. This bond has a put option meaning, if the value declines below a certain threshold, it will be sold at a specific price. We incorporated this by having every bond valued below $100 in year 3 be valued at $100, Every bond valued below $102.50 in year 2 being valued at $102.5, and every bond valued under $105 in year 1 being valued at $105. To get the values, we used the average price of two connecting nodes in a sequential year divided by the current forward rate.

Question 1c.)-We computed the OAS of the puttable bond by adding a fixed rate to all forward rates in all nodes of the chart. We then used "solver", setting the value of the bond in node 0 at the market price. We then set the variable to change at that fixed rate. The result gave us a OAS of 266 basis points.

Question 1.d)-We found the duration and convexity by recalibrating the binomial table according to the rate of shock. We both increased and decreased the yield to maturity by 20 basis points. Once we did that, we followed the same process that we had done in question 1a.) and recalibrated the tree for both the negative and positive shock rate. We

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then added the OAS from question 1c.) to every forward rate in both charts to get the negative shock and positive shock prices. We then solved the effective duration by subtracting the positive shock from the negative shock and dividing the original price multiplied by two times the absolute value of the shock rate. We then solved convexity by adding the positive and negative shock, subtracting the sum by two times the original price, and dividing the original price multiplied by the shock rate squared.

Question 2 – Binomial Tree and embedded options

Question 2a.)-We solved the question by plugging in the coupon value to all nodes excluding node zero. We then set year 4 values at par and plugged in the given forward rates. As previously mentioned, we found the forward rates for the higher nodes by using the formula e^(n*2*volatility). We then found the values of each node by dividing the average of the two sequential nodes of a node by the node's forward rate. If there is any value in year 3 higher than 100, we value it as 100. If there is any value in year 2 higher than 100.5, we value it as 100.5, and if there is any value in year 1 higher than 101, we value it at 101.

Question 2b.)-For question 2b.) We used a very similar format to question 2b.). A primary difference we had was that every year we increased the coupon value by 100 bps/1%. We then solved it in the same manner as we did 2b.) Where the values of the bonds were the minimum between the call value and the current value.

Question 2c.)- In question 2c.), we found the value of the coupons by adding 95 basis points to the forward rate value. We then valued the coupon as being the maximum between the coupon value and 4, and the minimum between the coupon value and 7. Question 2c.) differs from the other binomial tables we have created by changing the formula for the average value of the sequential nodes. In previous questions, the formula was 0.5*((“sequential value” + “sequential coupon”))/(1+”current forward rate”)). This question differs as we used the current coupon rate to calculate the average value of the sequential nodes. Once we had implemented this change, everything else remained consistent with how we solved the price in the prior questions.

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Question 3 – Passthrough and CMO

Question 3a.) -As a first step, we compute the monthly payments on the pooled mortgages with a total balance of $350 million by using Excel's PMT function for each month until maturity, applying the corresponding outstanding balance each time (e.g., =PMT(r = 7%, nper = 24 – (n – 1), PV = –350M, FV = 0, 0)).

The net interest received by the investor is simply the difference between the WAC paid by the mortgages and the servicing fee charged by the bank (7% – 1% = 6% net interest). This net rate is then multiplied by the outstanding balance for each corresponding month.

To compute the total principal payment, we must account for several components, including scheduled principal and prepayments:

Scheduled principal:
To compute this value in our amortization table, we subtract the mortgage interest for the period (WAC × outstanding balance for month n) from the total mortgage payment calculated previously.

Prepayment:
To compute this value, we must account for the Single Monthly Mortality (SMM) rate, which is determined using the following formula:

SMM = 1-(1-CPR) ^1/12, Where **CPR = (t/30)*6%**

For this example, we have a PSA of 200, which means the CPR must be multiplied by 200/100 (i.e., CPR × 2).

Once the SMM is computed, we can calculate the prepayment amount using:
Prepayment = (Outstanding Balance_n – Scheduled Principal_n) × SMM_n.

Finally, to obtain the total principal, you subtract the prepayment amount from the scheduled principal.

As the last step, to calculate the total cash flow delivered to investors, you add the net interest to the total principal.

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Question 3b.) -To calculate the Macaulay Duration of the passthrough, we first compute its price by discounting the cash flows at a rate of 4.80%, which reflects similar risk.

We use the following discount factor:
$1 / (1 + (4.80\% / 12))^{nper}$

The cash flow to be discounted for each period consists of the net interest plus the total principal received by the investor.

By applying this discount factor to every period's cash flow and summing up the results, we obtain the total price of the passthrough, which is $354,248,707.

Once the present value of all cash flows is computed, we can calculate the Macaulay Duration using the standard formula:

$$\text{Macaulay duration} = \frac{1}{P}\left[1 \times \frac{C}{(1+i)} + 2 \times \frac{C}{(1+i)^2} + \ldots + n \times \frac{(C+\text{Par})}{(1+i)^n}\right],$$

For this passthrough, we obtain a Macaulay duration of **1.012 years**, which is calculated by taking the sum of each period's **discounted cash flow multiplied by its corresponding period**, and then dividing that sum by the **price of the passthrough**.

Question 3c.) -Each of the tranches has their own outstanding balance and coupon rate which are applied to that balance to account for the net interest delivered to investors:

Tranches	Par amount	Cpn
A	$ 140,000,000	4.60%
B	$ 70,000,000	4.80%
C	$ 60,000,000	5.00%
D	$ 80,000,000	5.20%
IO		6.00%
Total	$ 350,000,000	Collateral

For **Tranche A**, the net interest is calculated by multiplying its coupon rate by its outstanding balance, which is $140 million. Because Tranche A has the highest payment priority, all total principal cash flows are directed to Tranche A until it is fully paid off

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(up to month 10). To determine each period's outstanding balance, we subtract the total principal payment for that period from the previous balance.

When we reach period 10, Tranche A's outstanding balance becomes smaller than the scheduled total principal payment. In that case, only the amount needed to fully retire from Tranche A is taken from the total principal. Once Tranche A is paid off at month 10, the remaining portion of that period principal payment is allocated to **Tranche B**. This process continues sequentially until **Tranche D** is fully paid off as well.

Since the **IO tranche** receives only interest and no principal, it follows a different computation process. As a first step, we calculate the **interest rate spread** between the IO tranche and each of the other tranches.

Tranches	Par amount	Cpn	Excess interest
A	$ 140,000,000	4.60%	1.40%
B	$ 70,000,000	4.80%	1.20%
C	$ 60,000,000	5.00%	1.00%
D	$ 80,000,000	5.20%	0.80%
IO		6.00%	
Total	$ 350,000,000	Collateral	

Once these values are obtained, we sum up the product of each excess interest rate and its corresponding tranche balance, then divide that total by the IO's annual interest rate to determine the **nominal IO balance**.

Month 1	Excess interest	Tranche Balances	Iterest spread amount
A	1.40%	$ 140,000,000	$ 1,960,000.00
B	0.012	70000000	$ 840,000.00
C	0.01	60000000	$ 600,000.00
D	0.008	80000000	$ 640,000.00
IO	6%		$ 67,333,333.33

This calculation is repeated for each period over the life of the CMO, up until the final payment is made to Tranche D.

Lastly, the interest amount allocated to the IO tranche is computed by multiplying the previously calculated nominal balance by the IO's monthly interest rate.

Tranche IO	
Balance	Interest
$67,333,333.33	=X60*.06/12
$64,127,084.63	$320,635.42
$60,879,203.97	$304,396.02
$57,592,785.74	$287,963.93
$54,270,993.20	$271,354.97

To compute the **average life** for each tranche, we sum up the product of each payment period and the total principal paid to that tranche. To express the average life in years, we divide this result by **12**. The same process is applied to all tranches except the **IO**, which receives no principal.

Tranches	Par amount	Cpn	Excess interest	Avg Life
A	$ 140,000,000	4.60%	1.40%	0.45
B	$ 70,000,000	4.80%	1.20%	1.06
C	$ 60,000,000	5.00%	1.00%	1.41
D	$ 80,000,000	5.20%	0.80%	1.81
IO		6.00%		
Total	$ 350,000,000	Collateral		1.04687

Question 4 – Monte Carlo Simulation on a Tree

Question 4a.)-For question 4a.) we first implemented the spot rates we were given in question 2 into a binomial chart. Once we had completed that, we calculated the value in the terminal nodes. We did this by inserting the rates in the final nodes into the formula $100x[max(*f*(1,3);8%)-min(*f*(1,3);5%)]. F(1,3) is the one year forward rate at year 3, which was given, so we inserted it into the formula giving us the value of all four ending nodes. Once we had those values, we plugged them into the formula that we have been using to derive prior node values. That formula being 0.5*((node(1,1) value)/(1+node0 forward rate)) +0.5*((node(1,2)value)/(1+node0 forward rate)). From there we were able to get a European option value of 3.4783. We further confirmed this by establishing a path chart in which we multiplied the terminal value of the path by its likelihood of being chosen, then further multiplying it by the discount rate. To find the year 3 discount rates we used 1/((fr(0,1)*fr(1,1)*fr(1,2)). The values of the forward rates would change depending on the path, giving us different discount rates.

Question 4b.)- We developed a Monte Carlo simulation by developing a table of all the paths and the rates that the paths consist of. In the Random column, we inserted the function Randomvalue(1,8) as to randomly choose a path between 1 and 8. Using VLOOKUP we then inserted the corresponding rates derived from the priorly mentioned table to the random variable. We then calculated the payoff by using the value at the terminal node of the path and multiplying it by the discount factor. We then did this for 100 random variables, summed up the average and divided it by 100 to get the price. We then found the standard deviation of the discounted cash flow and divided it by 100 squared to get the standard error. We used the standard error and z value of 1.959964 to find the high and low levels of the Monte Carlo simulation. For this, we added and subtracted the formula (z-value*standard error) to the Monte Carlo formula. Lastly, we subtracted the Monte Carlo Price from the true price to give us our difference in values.

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